Question:

You are assigned to configure a **Network File Sharing system using NFS** between two Linux machines.

Server Requirements (192.168.220.130):

1. Install the required NFS package.
2. Create a shared directory /data.
3. Create a file named secret inside /data.
4. Set appropriate permissions so all users can read and execute files.
5. Configure NFS export such that:
    - Only network 192.168.220.0/24 can access it
    - Access permissions: **read-write**
    - Changes must be **synchronous**
    - Root user on client should have full access
6. Start and enable NFS service.
7. Configure firewall to allow NFS traffic.

Client Requirements (Any client machine):

1. Install required NFS client package.
2. Create a mount point /mnt.
3. Mount the NFS share **temporarily**.
4. Verify the mounted filesystem.
5. Unmount it.
6. Configure **permanent mounting** using /etc/fstab.
7. Mount all filesystems and verify again.

Verification Tasks:

1. Confirm share is mounted using appropriate command.
2. Access the file secret from client.
3. Verify root access behavior (due to no_root_squash).
4. Check disk usage of mounted share.

--x--

```
# ==========================================================
# NFS COMPLETE SETUP (SERVER + CLIENT + VERIFICATION)
# ==========================================================



# ==========================================================
# SERVER CONFIGURATION (192.168.220.130)
# ==========================================================

# 1) Install NFS
dnf install -y nfs-utils

# 2) Create shared directory
mkdir -p /data

# 3) Create file
touch /data/secret

# 4) Set permissions
chmod -R 755 /data

# 5) Configure exports
vim /etc/exports

# Add:
# /data   192.168.220.0/24(rw,sync,no_root_squash)

# Apply export config
exportfs -rav
```

```
# 6) Start and enable services
systemctl enable --now rpcbind
systemctl enable --now nfs-server

# Verify service
systemctl status nfs-server

# 7) Configure firewall
firewall-cmd --add-service=nfs --permanent
firewall-cmd --reload

# Verify export
showmount -e 192.168.220.130



# =========================================================
#  CLIENT CONFIGURATION
# =========================================================

# 1) Install NFS client
dnf install -y nfs-utils

# 2) Create mount point
mkdir -p /mnt

# 3) Temporary mount
mount -t nfs 192.168.220.130:/data /mnt
```

```
# 4) Verify mount
df -h
mount | grep nfs

# 5) Unmount
umount /mnt

# 6) Permanent mount
vim /etc/fstab

# Add:
# 192.168.220.130:/data   /mnt   nfs   defaults   0 0

# 7) Apply fstab
mount -a

# Verify again
df -h



# ==========================================================
#  VERIFICATION TASKS
# ==========================================================

# 1) Confirm mount
df -h | grep /mnt

# 2) Access file
ls /mnt
```

```
cat /mnt/secret

# 3) Root access test (no_root_squash)
touch /mnt/root_test
chown root:root /mnt/root_test
ls -l /mnt

# Expected:
# root_test owned by root → confirms no_root_squash

# 4) Disk usage
df -h /mnt
du -sh /mnt



# ==========================================================
# 🔧 TROUBLESHOOTING (EXAM QUICK FIX)
# ==========================================================

# Check server export
showmount -e 192.168.220.130

# Restart services
systemctl restart nfs-server
systemctl restart rpcbind

# Re-export
exportfs -rav
```

```
# Firewall check
firewall-cmd --list-all

# Check mount errors
mount -a

# Check network
ping 192.168.220.130
```

```
# "NFS allows sharing directories over network.
# Server exports using /etc/exports, client mounts using mount or fstab,
# and verification is done using df, ls, and permission testing."
```

---x---

You are assigned to configure a Network File Sharing system using NFS between two Linux machines.

Server Requirements (192.168.220.130):

1. Install the required NFS package.

```
root ~
dnf install -y nfs-utils
```

2. Create a shared directory /data.

3. Create a file named secret inside /data.

```
root ~
mkdir /data && touch /data/secret.txt
```

4. Set appropriate permissions so all users can read and execute files.

```
root ~
chmod +x /data/secret.txt
root ~
ls -l /data/secret.txt
-rwxr-xr-x. 1 root root 0 Apr 11 23:24 /data/secret.txt
```

5. Configure NFS export such that:

- Only network 192.168.220.0/24 can access it
- Access permissions: read-write
- Changes must be synchronous
- Root user on client should have full access

```
root@server200:~ – vim /etc/exports
root@server200:~ – vim /etc/exports        root@server200:~

#########################################################
/data           192.168.220.0/24(rw,sync,no_root_squash)
#########################################################

#/data    192.168.1.254/32(rw,sync,no_root_squash)
#/data          *(ro,sync,no_root_squash)
# =========================================================
# NFS EXPORT CONFIGURATION FILE
# File: /etc/exports
# Purpose: Define which directory is shared, to whom, and with what permissions
# =========================================================


# =========================================================
# MAIN CONFIGURATION (LAB STANDARD)
# Share directory: /data
# Allow ONLY network 192.168.220.0/24
# Permissions: Read + Write
# =========================================================
```

5. Start and enable NFS service.

```
root ~
🐧 systemctl restart nfs-utils
```

7. Configure firewall to allow NFS traffic.

```
_____------Allowing NFS on firewall----------____
root ~
🐧 firewall-cmd --add-service=nfs --permanent
success
root ~
🐧 firewall-cmd --reload
success
root ~
🐧 firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client ftp http https nfs ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

Client Requirements (Any client machine):

1. Install the required NFS client package.

```
root@localhost:~# dnf install-y nfs-utils
```

2. Create a mount point /mnt.

3. Mount the NFS share temporarily.

```
root@localhost:~# mount -t nfs 192.168.220.130:/data /mnt
```

4. Verify the mounted filesystem.

```
root@localhost:~# ls /mnt
secret.txt
root@localhost:~# df -h
Filesystem             Size  Used Avail Use% Mounted on
/dev/mapper/cs-root    7.0G  4.4G  2.6G  63% /
devtmpfs               1.2G     0  1.2G   0% /dev
tmpfs                  1.2G   84K  1.2G   1% /dev/shm
tmpfs                  463M  9.1M  454M   2% /run
tmpfs                  1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
/dev/nvme0n1p2         4.9G  503M  4.4G  11% /boot
tmpfs                  232M  124K  232M   1% /run/user/0
/dev/sr0               9.5G  9.5G     0 100% /run/media/root/CentOS-Stream-10-BaseOS-x
86_64
192.168.220.130:/data  7.0G  4.5G  2.5G  65% /mnt
root@localhost:~# S
```

6. Unmount it.

```
root@localhost:~# umount /mnt
root@localhost:~# df -h
Filesystem           Size  Used Avail Use% Mounted on
/dev/mapper/cs-root  7.0G  4.4G  2.6G  63% /
devtmpfs             1.2G     0  1.2G   0% /dev
tmpfs                1.2G   84K  1.2G   1% /dev/shm
tmpfs                463M  9.1M  454M   2% /run
tmpfs                1.0M     0  1.0M   0% /run/credentials/system
-journald.service
/dev/nvme0n1p2       4.9G  503M  4.4G  11% /boot
tmpfs                232M  124K  232M   1% /run/user/0
/dev/sr0             9.5G  9.5G     0 100% /run/media/root/CentOS-
tream-10-BaseOS-x86_64
root@localhost:~# ls /mnt
hgfs
```

7. Configure permanent mounting using /etc/fstab.

```
root@localhost:~ – vim /etc/fstab
root@localhost:~    root@localhost:~ – vim /etc/fstab
# /etc/fstab
# Created by anaconda on Thu Apr  9 15:33:24 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=a74dce93-fb34-436e-afb4-71ecc4d9ba1e /
    xfs     defaults        0 0
UUID=1bb1cbd2-dad0-4f61-a776-0edb7c55c91f /boot
    xfs     defaults        0 0
UUID=f392b621-d550-435f-b002-613dd1837886 none
    swap    defaults        0 0


#===================================================

192.168.220.130:/data   /mnt    nfs defaults 0  0

```

7. Mount all filesystems and verify again.

```
root@localhost:~# vim /etc/fstab
root@localhost:~# systemctl daemon-reload
root@localhost:~# mount -a
root@localhost:~# ls /mnt
secret.txt
root@localhost:~# mount | grep nfs
192.168.220.130:/data on /mnt type nfs4 (rw,relatime,vers=4.2,rsize
=262144,wsize=262144,namlen=255,hard,fatal_neterrors=none,proto=tcp
,timeo=600,retrans=2,sec=sys,clientaddr=192.168.220.131,local_lock=
none,addr=192.168.220.130)
root@localhost:~# df -h
Filesystem             Size  Used Avail Use% Mounted on
/dev/mapper/cs-root    7.0G  4.4G  2.6G  63% /
devtmpfs               1.2G     0  1.2G   0% /dev
tmpfs                  1.2G   84K  1.2G   1% /dev/shm
tmpfs                  463M  9.1M  454M   2% /run
tmpfs                  1.0M     0  1.0M   0% /run/credentials/syste
md-journald.service
/dev/nvme0n1p2         4.9G  503M  4.4G  11% /boot
tmpfs                  232M  124K  232M   1% /run/user/0
/dev/sr0               9.5G  9.5G     0 100% /run/media/root/CentOS
-Stream-10-BaseOS-x86_64
192.168.220.130:/data  7.0G  4.5G  2.5G  65% /mnt
root@localhost:~# touch /mnt/testfile
root@localhost:~# ls /mnt
secret.txt  testfile
```

Verification Tasks:

1. Confirm share is mounted using appropriate command.
2. Access the file secret from client.
3. Verify root access behaviour (due to no_root_squash).
4. Check disk usage of the mounted share.